

Homework 3

- 1 Consider the Schwarzschild black hole with $G = 1$,

$$ds^2 = -\left(1 - \frac{2M}{r}\right)dt^2 + \frac{dr^2}{1 - \frac{2M}{r}} + r^2(d\theta^2 + \sin^2\theta d\phi^2). \quad (1)$$

- a Check that in the weak field limit ($r \gg 2M$),

$$g_{tt} = -(1 + 2\Phi), \quad g_{rr} = 1 - 2\Phi, \quad (2)$$

where $\Phi = -\frac{M}{r}$ is the Newton's potential.

- b Work out the Killing vectors of the Schwarzschild solution and check the algebra generated by them.
c Use the tetrad formalism to calculate R_{ab} by hand.
d Use Mathematica to calculate R_{ab} . Alternative packages: RGTC, diffgeo, xAct...
e Check that the Kretschmann scalar $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ diverges at $r = 0$, but not at the Schwarzschild radius $r_s = 2M$.

- 2 Consider a geodesic congruence $x^\mu(\tau, s)$ generated by u^a ,

- a Derive the geodesic deviation equation

$$a^\mu = \frac{D^2}{d\tau^2}\eta^\mu = R^\mu_{\nu\lambda\sigma}u^\nu u^\lambda \eta^\sigma \quad (3)$$

where a^μ is the 4-acceleration, $u^\mu = \frac{\partial x^\mu}{\partial \tau}$ is the tangent vector field to the geodesic congruence, $\eta^\mu = \frac{\partial x^\mu}{\partial s}$ is the deviation (separation) vector pointing from one geodesic to a neighboring one.

- b For a free falling observer in the Schwarzschild BH, calculate the tidal force (the 4-acceleration a^μ),

$$\frac{D^2}{d\tau^2}\eta^1 = \frac{2M}{r^3}\eta^1, \quad \frac{D^2}{d\tau^2}\eta^2 = -\frac{M}{r^3}\eta^2, \quad \frac{D^2}{d\tau^2}\eta^3 = -\frac{M}{r^3}\eta^3. \quad (4)$$

The plus sign means tension or stretching in the radial direction, while the minus sign means pressure or compression or squashing

along the transversal directions. The instantaneous LIF of the free falling observer is taken as

$$\hat{e}^0 = dT, \quad \hat{e}^1 = dr + \sqrt{\frac{2M}{r}}dT, \quad \hat{e}^2 = r d\theta, \quad \hat{e}^3 = r \sin \theta d\phi. \quad (5)$$

where T is the proper time of the free falling observer moving radially from infinity. The Painleve-Gullstrand coordinate reads

$$ds^2 = -dT^2 + (dr + \sqrt{\frac{2M}{r}}dT)^2 + r^2 d\Omega_2^2. \quad (6)$$